

ISRO NOC COPILOT - NETWORK TOPOLOGY AND DEVICE INVENTORY

1. NETWORK OVERVIEW

The network consists of a dual-tier MPLS core with SD-WAN overlay for branch connectivity. The backbone connects 4 Points of Presence (PoPs): Delhi (primary hub), Mumbai, Bangalore, and Hyderabad. Each PoP contains redundant PE routers. 38 branch sites connect via SD-WAN WAN Edge routers using dual ISP links (BSNL MPLS primary, Airtel broadband secondary).

2. CORE ROUTERS - PE DEVICES

PE-DEL-01 | Cisco ASR 9001 | Delhi DC Primary | 192.168.10.1 | AS 65001
PE-DEL-02 | Cisco ASR 9001 | Delhi DC Secondary | 192.168.10.2 | AS 65001
PE-MUM-01 | Cisco ASR 9006 | Mumbai PoP | 192.168.20.1 | AS 65001
PE-MUM-02 | Cisco ASR 9006 | Mumbai PoP Backup | 192.168.20.2 | AS 65001
PE-BLR-01 | Cisco ASR 9001 | Bangalore PoP | 192.168.30.1 | AS 65001
PE-HYD-01 | Cisco ASR 9001 | Hyderabad PoP | 192.168.40.1 | AS 65001

All PE routers run: IOS-XR 7.9.2 | BGP-4 | MPLS LDP + RSVP-TE | IS-IS Level-2

3. SD-WAN CONTROLLERS

vManage-01 (Primary) | 10.0.0.10 | vManage 20.9.3 | Delhi DC
vManage-02 (Secondary) | 10.0.0.11 | vManage 20.9.3 | Mumbai DC
vSmart-01 | 10.0.0.20 | vSmart 20.9.3 | Delhi DC
vSmart-02 | 10.0.0.21 | vSmart 20.9.3 | Bangalore DC
vBond-01 | 10.0.0.30 | vBond 20.9.3 | Public DMZ

Certificate Authority : Cisco PKI | Renewal cycle: Every 12 months

vManage admin URL : https://10.0.0.10:443

4. BRANCH WAN EDGE DEVICES

Branch naming convention: WE-<CITY>-<INDEX>

WE-AGR-01 | Cisco Catalyst 8200 | Agra Branch | 10.1.1.1 | BSNL+Airtel
WE-LKO-01 | Cisco Catalyst 8200 | Lucknow Branch | 10.1.2.1 | BSNL+Jio
WE-JDP-01 | Cisco Catalyst 8300 | Jodhpur Branch | 10.1.3.1 | BSNL+Airtel
WE-PUN-01 | Cisco Catalyst 8300 | Pune Branch | 10.1.4.1 | BSNL+Airtel
WE-PUN-02 | Cisco Catalyst 8200 | Pune Branch-2 | 10.1.4.2 | BSNL+Airtel
WE-CHN-01 | Cisco Catalyst 8300 | Chennai Branch | 10.1.5.1 | BSNL+ACT
WE-KOC-01 | Cisco Catalyst 8200 | Kochi Branch | 10.1.6.1 | BSNL+Airtel
[... 31 additional branch sites follow same naming convention]

5. MPLS BACKBONE LINKS

Delhi -> Mumbai : 10 Gbps STM-64 | Carrier: BSNL Leased | SLA: 99.99%
Delhi -> Bangalore : 10 Gbps STM-64 | Carrier: Tata Comm | SLA: 99.99%
Delhi -> Hyderabad : 1 Gbps STM-16 | Carrier: BSNL Leased | SLA: 99.95%
Mumbai -> Bangalore : 10 Gbps STM-64 | Carrier: Reliance | SLA: 99.99%
Mumbai -> Hyderabad : 1 Gbps STM-16 | Carrier: BSNL Leased | SLA: 99.95%
Bangalore -> Hyderabad : 1 Gbps STM-16 | Carrier: Tata Comm | SLA: 99.95%

All backbone links are protected by RSVP-TE Fast Reroute (FRR) with 50ms convergence.

6. IP ADDRESS PLAN

Loopback addresses : 192.168.0.0/24 (all PE/P routers)

MPLS backbone links : 172.16.0.0/16

Branch LAN subnets : 10.0.0.0/8 (each branch /24)

SD-WAN overlay : 100.64.0.0/10 (RFC6598 carrier-grade NAT range)

Management network : 10.255.0.0/24 (out-of-band)

NMS servers : 10.255.0.10-20

IPAM Tool : phpIPAM at <https://ipam.noc.internal>

7. CRITICAL DEPENDENCIES

DNS Servers : 10.255.0.10 (primary), 10.255.0.11 (secondary)

NTP Servers : 10.255.0.5 (Stratum-1, GPS-synced)

TACACS+ : 10.255.0.15 (authentication for all network devices)

Syslog Server : 10.255.0.16 (ELK Stack)

SNMP Community : noc-monitor-2024 (read-only)

Zabbix NMS : <https://nms.noc.internal>

NetFlow Collector: 10.255.0.17 (Grafana dashboard)

WARNING: If TACACS+ server is unreachable, use local fallback account.

Local credentials are stored in sealed envelope in NOC safe.